

Lombardia Vivibile

Building a Multidimensional Quality of Life Index
for Lombardy's Main Cities

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1. Introduction

This report presents the project "Lombardia Vivibile". The project builds a composite liveability index (0-100) for five major Lombard cities (Milan, Brescia, Bergamo, Monza, and Como) by integrating heterogeneous data from publicly available sources through a complete data management pipeline.

The project was motivated by a key observation: liveability is inherently multidimensional. A city with many parks may still have poor air quality; a city with extensive public transport may score poorly on cultural services. Standard rankings typically aggregate these dimensions in a simplistic way. Our approach instead aims to make the weights and trade-offs explicit and methodologically transparent.

The core research goal is not merely to produce a ranking, but to identify the drivers that differentiate liveability across cities, particularly by investigating the relationship between urban green spaces and air quality through spatial data enrichment.

1.1 Research Questions

The project is structured around four research questions:

1. **RQ1 - Ranking:** Which Lombard city ranks highest on a multidimensional liveability index that integrates urban green spaces, cultural services, public transport, and air quality?
2. **RQ2 - Drivers:** Which dimensions contribute most to liveability? Is there a dominant dimension that differentiates cities more than the others?
3. **RQ3 - Usability:** Are urban parks located in areas with acceptable air quality according to WHO/EU limits? Are green spaces and air quality truly independent dimensions of liveability?
4. **RQ4 - Data Quality:** Are the datasets collected from heterogeneous sources complete, consistent, and reliable enough for a quantitative comparison across cities?

2. Data Sources and Acquisition Techniques

All data used in this project are publicly and freely available. Three sources were selected to cover the main dimensions of liveability, acquired through two distinct techniques.

Source	Technique	Data Acquired	Documents
OpenStreetMap Overpass API	REST API	Green areas, cultural POIs, bus stops	8,625
ARPA Lombardia	CSV Download	NO2, Ozone, Benzene measurements	3,684
ISTAT demo.istat.it	ZIP Download	Population by age group and gender	510

2.1 OpenStreetMap - Overpass API

OpenStreetMap was queried via the Overpass API, an HTTP REST interface that allows complex spatial queries on OSM data. For each of the 5 cities, 3 queries were executed corresponding to 3 categories of points of interest:

- Green spaces: leisure ~ "park|garden|nature_reserve"
- Cultural services: amenity ~ "theatre|museum|library|cinema"
- Public transport: "highway = bus_stop"

The search radius was adapted to the size of each city: 8,000m for Milan, 5,000m for Brescia, 4,000m for Bergamo, and 3,500m for Monza and Como. Rate limiting was managed with a 2-second pause between requests and a mandatory User-Agent header. A total of 8,625 documents were collected.

It should be noted that OSM data completeness varies across cities depending on the level of community mapping activity. Smaller or more tourist-oriented cities such as Como tend to have less comprehensive coverage compared to larger urban centres. This potential bias is acknowledged as a methodological limitation and is discussed further in Section 10.1

2.2 ARPA Lombardia - CSV Download

Air quality data was obtained from the ARPA Lombardia open data portal. Two CSV files were downloaded: a measurements dataset (50,000 rows) and a station registry (1,002 stations). The two datasets were joined on the `idsensore` field and filtered to retain only active stations in the 5 target cities measuring NO2, Ozone, and Benzene. The final dataset contains 3,684 valid measurements across 29 active stations.

Measurements were excluded from the final dataset for three reasons:

- the station was no longer active (`storico = 'Y'`), meaning it did not provide current data;
- the station was located outside the five target cities, as ARPA Lombardia monitors air quality across the entire region;
- the station measured pollutants other than NO2, Ozone, and Benzene, which were the only indicators selected for the liveability index due to their relevance to urban traffic emissions and the availability of objective EU/WHO reference limits.

2.3 ISTAT - ZIP Download

Population data was obtained from ISTAT's demo.istat.it portal, which provides downloadable archives of resident population statistics per municipality. Five ZIP files were downloaded, one per province (Como 013, Milan 015, Bergamo 016, Brescia 017, Monza and Brianza 108).

Each ZIP contained a CSV file with semicolon (;) as separator and a header row to skip. Data was filtered using the official ISTAT municipality codes for the five capitals: 013075 (Como), 015146 (Milan), 016024 (Bergamo), 017029 (Brescia), 108033 (Monza). The resulting dataset contains 510 records with population broken down by age group (0 to 100+) and gender.

3. Software Architecture

The project is developed entirely in Python 3.11 on JupyterLab, running inside a Docker container on an Azure Lab Services virtual machine.

MongoDB 5.0 is used as the primary NoSQL storage layer.

3.1 Technology Stack

- JupyterLab - development environment (Docker container on Azure VM)
- Python 3.11 - main programming language
- MongoDB 5.0 - NoSQL document database (Docker container)
- Mongo Express - web-based MongoDB UI for visual inspection
- Key Python libraries: pymongo, pandas, numpy, requests, matplotlib, zipfile, math

3.2 MongoDB Collections

Collection	Documents	Description
osm_raw	8,645	OSM POIs with coordinates and tags, enriched with nearest ARPA station air quality
arpa_raw	3,684	Air quality measurements filtered for the 5 cities
istat_raw	510	Population by age group and gender per municipality
indice_vivibilita	5	Final liveability index score per dimension and per city
osm_arricchito_summary	15	Average air quality aggregated by POI category and city

3.3 Notebook Structure

Notebook	Description
01_acquisizione_osm.ipynb	OSM data acquisition via Overpass API
02_acquisizione_arpa.ipynb	ARPA CSV download, join with station registry
03_acquisizione_istat.ipynb	ISTAT ZIP download, parsing, filtering
04_integrazione.ipynb	Dataset join, index computation
05_profiling.ipynb	Data quality assessment
06_analisi_esplorativa.ipynb	Exploratory analysis, visualisations, MongoDB queries
07_arricchimento_spaziale.ipynb	Spatial enrichment via Haversine distance

4. Data Management Pipeline

The pipeline follows a structured sequence of 6 phases, from raw data acquisition to the final analytical output.

4.1 Acquisition

Data is collected from the 3 sources using 2 different techniques. The Overpass API is queried programmatically with POST requests, while ARPA and ISTAT data are downloaded as structured files. All raw data is immediately persisted to MongoDB as-is, preserving the original structure for reproducibility.

4.2 Storage

Each document stored in MongoDB includes standard metadata fields: *citta* (city name), *categoria* (category), *acquired_at* (UTC timestamp), and source-specific fields such as lat/lon coordinates, OSM tags, or measurement values.

4.3 Data Profiling and Quality Assessment

Before integration, all collections were systematically profiled for completeness, consistency, outlier detection, and cross-source validation. Results are detailed in Section 5.

4.4 Integration

The three datasets are integrated at the city level via an inner join on the city name field. OSM counts are normalised by city surface area (km²) using a logarithmic scale. Air quality data is aggregated as a mean per city and pollutant, filtering only valid measurements.

4.5 Spatial Enrichment

Every OSM POI was enriched with the air quality of the nearest ARPA monitoring station using the Haversine formula. This is described in detail in Section 6.

4.6 Index Computation and Analysis

The composite liveability index is computed using a weighted sum of 5 normalised dimensions. Air quality scores are derived from absolute WHO/EU benchmarks. The final scores are persisted to the *indice_vivibilita* collection and visualised through three charts.

5. Data Quality Assessment

5.1 Completeness

Completeness was measured as the percentage of null values on key numeric fields across all collections:

- osm_raw: 0% null on lat, lon fields
- arpa_raw: 0% null on valore, lat, lng fields
- istat_raw: 0% null on totale field

All collections achieved 100% completeness on their primary numeric fields.

5.2 Consistency

Of the 3,684 ARPA measurements collected, 3,677 (99.8%) have status VA (valid). The remaining 7 measurements have status NaN and were excluded from air quality aggregations.

5.3 Outlier Detection

Outliers were detected using the threshold of mean + 3 standard deviations per pollutant:

Pollutant	Outliers	Threshold (ug/m3)	Note
Benzene	17	1.87	Likely seasonal peaks
Ozone	8	156.85	Expected summer peaks
NO2	13	69.12	Traffic-related spikes

All 38 outliers were retained in the dataset as they represent physically plausible events rather than data errors.

5.4 ISTAT Population Anomaly

This section documents a data quality issue identified during profiling, the corrective action taken, and post-correction validation.

During profiling, population totals aggregated from istat_raw were compared against official ISTAT 2023 Census figures. An initial discrepancy of approximately 50% was observed for Milan. Investigation revealed the cause: the ISTAT dataset reports males and females as separate rows for each age group, so direct summation was double-counting the population. The fix was to divide the aggregated total by 2.

Post-correction validation:

City	DB (corrected)	Official 2023	Delta
Milan	1,362,863	1,371,498	-0.6%
Brescia	201,342	197,167	+2.1%
Bergamo	120,629	121,738	-0.9%
Monza	123,672	123,196	+0.4%
Como	83,035	83,480	-0.5%

Post-correction deltas are below 2.1% for all cities, confirming good underlying data quality.

6. Spatial Enrichment and Integration Quality

The spatial enrichment phase links OSM points of interest with the air quality of their surrounding area as measured by the nearest ARPA monitoring station.

6.1 Method

For every OSM document with valid lat/lon coordinates, the nearest ARPA monitoring station in the same city was identified using the Haversine formula:

```
d = 2R * arcsin( sqrt( sin2((lat2-lat1)/2) + cos(lat1)*cos(lat2)*sin2((lon2-lon1)/2) ) )
```

where R = 6,371 km. The station with the minimum distance was selected and its air quality metrics (mean NO₂ and mean Ozone) were written back to the OSM document via a MongoDB update_one operation.

6.2 Integration Coverage and Quality Metrics

The following metrics characterise the quality of the spatial enrichment:

Metric	Value
Total POIs processed	8,645
POIs successfully enriched	8,645 (100%)
POIs without a matching station	0 (0%)
Active ARPA stations with coordinates	13
Estimated average distance POI to station	~3-4 km
Maximum observed distance	~7 km (Milan Verziere example: 6.6 km)

Note: distance statistics are estimated based on OSM query radii (3,500-8,000m) and documented examples. Exact aggregate figures are computed in notebook 07_arricchimento_spaziale.ipynb and can be reproduced by running the full pipeline.

The 100% enrichment rate is achieved because the ARPA station coverage within each city is sufficient to assign a nearest station to every OSM POI within the query radius. No fallback to inter-city stations was required.

6.3 Enrichment Results

The enrichment was aggregated by city and POI category to compute the mean air quality in areas surrounding each type of urban service. The most significant finding concerns parks:

City	Category	NO2 area (ug/m3)	vs EU limit (40)	Quality
Bergamo	Parks	21.1	-47%	GOOD
Brescia	Parks	21.2	-47%	GOOD
Como	Parks	26.4	-34%	GOOD
Monza	Parks	34.5	-14%	GOOD
Milan	Parks	56.4	+41%	CRITICAL

Key insight: Milan has the highest number of green areas in absolute terms (2,184 OSM elements), yet its parks are located in areas where NO2 is 56.4 ug/m3 on average - 41% above the EU annual limit of 40 ug/m3. This demonstrates that green space quantity and air quality are not independent dimensions of liveability.

7. MongoDB Queries

As required by the course, at least two queries were implemented on the stored data. Three demonstrative queries are presented here, covering different aspects of the dataset.

7.1 Query 1 - Cities Exceeding the EU NO2 Limit

This query identifies cities where the average NO2 concentration exceeds the EU annual limit of 40 ug/m3, ordering by value descending. It demonstrates the use of MongoDB aggregation pipelines with multi-stage filtering.

```
db.arpas_raw.aggregate([
  { $match: { inquinante: "Biossido di Azoto", stato: "VA" } },
  { $group: { _id: "$citta", media_no2: { $avg: "$valore" } } },
  { $match: { media_no2: { $gt: 40 } } },
  { $sort: { media_no2: -1 } }
])
```

Result: Milan (50.27 ug/m3) is the only city exceeding the EU limit among the five analysed.

7.2 Query 2 - POI Count by Category and City

This query counts points of interest per category and city, ordered by total count descending. It provides an overview of the distribution of urban services across the dataset.

```
db.osm_raw.aggregate([
  { $group: {
    _id: { citta: "$citta", categoria: "$categoria" },
    totale: { $sum: 1 }
  } },
  { $sort: { totale: -1 } },
  { $limit: 10 }
])
```

Top results: Milan transport (3,070 POIs), Milan green (2,184 POIs), Brescia transport (894 POIs).

7.3 Query 3 - Final Liveability Ranking

This query retrieves the final ranking from the `indice_vivibilita` collection, ordered by rank. It demonstrates a simple document retrieval with projection and sorting.

```
db.indice_vivibilita.find(  
  {},  
  { _id: 0, rank: 1, citta: 1, indice_vivibilita: 1, score_aria: 1 }  
) .sort({ rank: 1 })
```

Result: Bergamo (70.5), Milan (67.5), Monza (66.6), Brescia (48.4), Como (26.9).

8. Liveability Index

8.1 Dimensions and Weights

Dimension	Weight	Metric	Normalisation
Urban green spaces	20%	Green areas / km2	Logarithmic scale
Cultural services	20%	Cultural POIs / km2	Logarithmic scale
Public transport	20%	Bus stops / km2	Logarithmic scale
Air quality - NO2	25%	Mean NO2 (ug/m3)	EU limits: 10-40 ug/m3
Air quality - Ozone	15%	Mean Ozone (ug/m3)	EU limits: 60-180 ug/m3

8.2 Methodological Choices

Normalisation by surface area (km2) instead of population

Urban services such as museums, parks, and bus stops serve the entire city regardless of population density. Normalising by surface area reflects the spatial accessibility of services rather than their per-capita availability.

Logarithmic scale for OSM counts

Raw OSM counts show extreme variation: Milan has 3,070 bus stops compared to Como's 231. A linear min-max normalisation would compress all cities other than Milan into a narrow range. The logarithmic transformation reduces this distortion while preserving ordinal rankings.

Absolute EU/WHO limits for air quality

With only 5 cities in the sample, relative min-max normalisation would assign a score of 100 to whichever city has the lowest pollution regardless of whether it is actually good in absolute terms. Using EU/WHO limits provides an objective external reference that is independent of the sample composition.

9. Results

9.1 Final Ranking

Rank	City	Green	Culture	Transport	Air	Index
1	Bergamo	79.0	36.4	76.5	80.4	70.5
2	Milan	100.0	63.7	100.0	36.9	67.5
3	Monza	62.0	100.0	69.2	50.9	66.6
4	Brescia	42.6	2.1	45.2	76.0	48.4
5	Como	0.0	0.0	0.0	67.3	26.9

9.2 Key Findings

Bergamo - 1st (70.5)

Bergamo ranks first primarily due to its air quality score of 80.4 - the highest among the five cities, with a mean NO₂ of 17.5 ug/m³, well below the EU limit. It also scores strongly on green spaces (79.0) and transport (76.5).

Milan - 2nd (67.5)

Milan dominates on urban services, achieving the maximum score (100.0) on both green spaces and public transport per km². However, it scores only 36.9 on air quality due to a mean NO₂ of 50.3 ug/m³ - the only city to exceed the EU annual limit of 40 ug/m³.

Monza - 3rd (66.6)

Monza achieves the maximum score on cultural services per km² (100.0), reflecting a high density of cultural POIs relative to its small surface area (33 km²).

Brescia - 4th (48.4)

Brescia shows a balanced profile with good air quality (76.0) but scores very poorly on cultural services (2.1), suggesting limited cultural infrastructure or incomplete OSM mapping.

Como - 5th (26.9)

Como ranks last due to near-zero scores on all OSM-derived dimensions. This is attributed to a methodological limitation: OSM data coverage appears significantly lower for Como than for the other cities, likely due to less complete community mapping for smaller tourist-oriented municipalities.

9.3 Air Quality Analysis

City	NO2 (ug/m3)	Ozone (ug/m3)	Note
Bergamo	17.5	73.6	Best NO2 - below EU guidance level
Brescia	21.2	65.2	Good air quality on both pollutants
Como	26.4	53.6	Best Ozone score - lowest in sample
Monza	34.5	55.6	Approaching EU NO2 limit
Milan	50.3	69.4	Exceeds EU NO2 annual limit (40 ug/m3)

All five cities exceed the WHO annual guidance value for NO2 (10 ug/m3). Milan is the only city to exceed the EU legal limit (40 ug/m3). All cities remain below both WHO and EU limits for Ozone.

10. Limitations and Future Work

10.1 Limitations

OSM Data Coverage for Como

Como consistently scores zero across all OSM-derived dimensions. The degree of underrepresentation (69 green areas vs 335 for Bergamo, comparable surface area) suggests incomplete community mapping rather than genuine absence of services. This limitation is documented rather than corrected to avoid introducing unsupported assumptions.

Sample Size

The analysis is based on 5 cities. The index weights and normalisation choices were tuned on this sample and may not generalise to a broader set of Lombard municipalities.

Index Weights

The weights (20/20/20/25/15) represent an informed but arbitrary choice not validated against expert opinion or citizen surveys.

ARPA Station Coverage

Two ARPA stations in Milan and Monza monitor only Ozone and not NO₂. POIs assigned to these stations lack an NO₂ value. For these cases, air quality metrics in the spatial enrichment were derived from Ozone readings, which are documented with a different reference scale.

10.2 Future Work

- Extension to all 12 Lombard provincial capitals
- Integration of official GTFS data from Regione Lombardia for a more robust transport metric
- Addition of a safety dimension using ISTAT crime statistics
- Automated periodic data refresh pipeline
- Weight sensitivity analysis to quantify the impact of different weighting schemes

11. Conclusions

This project built a composite liveability index for five Lombard cities by integrating data from three public sources through a complete data management pipeline: from API acquisition and file download to MongoDB storage, data profiling, spatial enrichment, and index computation.

5. **RQ1 - Ranking:** Bergamo ranks first (70.5), driven by excellent air quality. Milan ranks second (67.5) despite dominating on urban services, due to NO2 levels exceeding the EU limit.
6. **RQ2 - Drivers:** Air quality is the most discriminating dimension. Cities with worse air quality lose up to 40 index points relative to their service scores.
7. **RQ3 - Usability:** Milan's parks are located in areas where NO2 is 56.4 ug/m3 on average - 41% above the EU limit. Green spaces and air quality are demonstrably not independent dimensions of liveability.
8. **RQ4 - Data Quality:** All collections achieved 100% completeness on numeric fields. A double-counting anomaly in the ISTAT dataset was identified and corrected (post-correction delta < 2.1%). 99.8% of ARPA measurements were valid.

"Green spaces and air quality are not independent dimensions - where you live green, you don't always breathe clean."

12. Data Sources and References

Data Sources

- OpenStreetMap - Overpass API: <https://overpass-api.de/api/interpreter>
- ARPA Lombardia - Air quality measurements: <https://www.dati.lombardia.it/resource/nicp-bhqi.csv>
- ARPA Lombardia - Station registry: <https://www.dati.lombardia.it/resource/ib47-atvt.csv>
- ISTAT - Resident population: <https://demo.istat.it/app/?i=POS&l=it>

Reference Limits

- WHO Air Quality Guidelines (2021): NO₂ annual mean 10 ug/m³, Ozone peak season 100 ug/m³
- EU Ambient Air Quality Directive (2008/50/EC): NO₂ annual limit 40 ug/m³, Ozone target 120 ug/m³
- ISTAT Censimento Permanente 2023 - official municipal population figures

Tools and Libraries

- Python 3.11 - <https://www.python.org>
- MongoDB 5.0 - <https://www.mongodb.com>
- pymongo, pandas, numpy, requests, matplotlib
- OpenStreetMap - <https://www.openstreetmap.org>